

NORTHWIND TRADERS SALES ANALYTICS PROJECT

(Exploratory Data Analysis and Business Insights using Excel, SQL, and Power BI)

Independent Data Analytics Project

Submitted by

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Data Analytics Project Report

INTRODUCTION

1. Project Overview

The Northwind Traders Sales Analytics Project is a comprehensive data analysis and visualization initiative aimed at uncovering valuable business insights from the Northwind dataset. The dataset represents a fictitious wholesale company that imports and exports specialty foods worldwide.

The project follows a complete data pipeline approach:

- **Data Collection & Preparation:** CSV data files were imported into MySQL for structured storage and query execution.
- **Exploratory Data Analysis (EDA):** SQL queries were used to extract datasets, which were further analyzed in Excel using pivot tables and charts to answer exploratory questions.
- **Visualization & Reporting:** A Power BI report was created containing 15 visuals (one for each business question) and a consolidated interactive dashboard to summarize overall insights.

The project emphasizes actionable insights in sales performance, customer segmentation, inventory trends, and employee contributions, helping stakeholders make informed, data-driven decisions.

2. Objective of the Project

The primary objective of this project is to design an **interactive, user-friendly, and visually compelling reporting system** that provides a holistic view of Northwind Traders' operations. Specifically, the project aims to:

- Analyze sales trends across products, categories, regions, and customers.
- Segment customers based on purchase patterns and identify high-value clients.
- Evaluate employee performance and their contributions to sales growth.
- Study inventory and shipping patterns to improve operational efficiency.
- Measure profitability and discount impact on overall revenue.
- Provide a centralized Power BI dashboard enabling stakeholders to explore data dynamically and uncover insights quickly.

Ultimately, the report enables Northwind Traders to strengthen competitiveness in the wholesale market by fostering data-driven decision-making.

3. Tools & Technologies Used

- **Microsoft Excel**
 - Used for Exploratory Data Analysis (EDA).
 - Pivot tables and charts for initial trend exploration and business question answering.
- **SQL (Structured Query Language)**
 - Extracted and transformed raw data from multiple tables.
 - Executed queries for sales, customers, products, and employee insights.
- **MySQL Database**
 - Served as the central storage for CSV files.
 - Allowed structured querying, joins, and data integrity validation.
- **Power BI**
 - Designed a 15-page interactive report with individual Q&A visuals.
 - Built a summary dashboard consolidating insights across sales, customers, employees, suppliers, and inventory.
 - Implemented slicers, drill-through, and dynamic filtering for interactivity.

4. Dataset Description

The project uses the Northwind Traders dataset, a widely recognized sample database that simulates the sales operations of a global wholesale distributor of specialty foods. The dataset includes multiple interrelated tables representing customers, employees, products, orders, suppliers, and shippers, enabling comprehensive analysis of business processes.

Below is a structured overview of the key tables:

Customers Table

This table stores information about the company's customers. It includes fields for customer ID, company name, contact name, contact title, address, city, region, postal code, country, phone, and fax. It enables customer segmentation, geography-based sales distribution, and identification of top customers.

Employees Table

This table stores information about the company's employees. It includes fields for employee ID, last name, first name, title, title of courtesy, birth date, hire date, address, city, region, postal code, country, home phone, extension, photo, notes, reports to, and photo path. These are used to measure employee performance, sales handled by each employee, and reporting hierarchy.

Orders Table

This table stores information about the company's orders. It includes fields for order ID, customer ID, employee ID, order date, required date, shipped date, ship via, freight, ship name, ship address, ship city, ship region, ship postal code, and ship country. It supports order tracking, sales by time periods, freight cost analysis, and regional sales performance.

Order Details Table

This table stores detailed information about the items within each order. It includes fields for order ID, product ID, unit price, quantity, and discount. It forms the fact table for sales analysis; enables calculation of revenue, profit, and discount impact.

Products Table

This table stores information about the company's products. It includes fields for product ID, product name, supplier ID, category ID, quantity per unit, unit price, units in stock, units on order, reorder level, and whether the product is discontinued. It facilitates product-level sales performance, stock level monitoring, and category analysis.

Suppliers Table

This table stores information about the company's suppliers. It includes fields for supplier ID, company name, contact name, contact title, address, city, region, postal code, country, phone, fax, and home page. It helps assess supplier contributions, supplier-product relationships, and sourcing dependencies.

Shippers Table

This table stores information about the company's shipping companies. It includes fields for shipper ID, company name, and phone. It enables analysis of shipping performance, delivery timelines, and freight cost efficiency.

Categories Table

This table stores information about the product categories. It includes fields for category ID, category name, and description. It supports category-level sales and profitability analysis.

METHODOLOGY

1. Exploratory Data Analysis (Excel)

The project explored **15 key business questions** using SQL queries and Excel pivot tables/charts. Each question was designed to address a specific dimension of Northwind Traders' business (sales, customers, employees, suppliers, inventory, shipping).

Structure of Analysis (Q1–Q15)

For each question:

1. **Business Question:** Defines the problem to be solved.
2. **SQL Query:** Extracts the required dataset from MySQL.
3. **Excel EDA:** Data exported into Excel for pivot-based exploration.
4. **Visualization:** Pivot charts used for initial insights.

Example (Customer Clustering Based on Spend, Orders, and Categories):

- **Business Question:** Can we cluster customers based on total spend, order count, and preferred categories?
- **SQL Query:**

```
with customer_summary as (  
    select c.CustomerID, count(distinct o.OrderID) as Order_Count, sum(d.UnitPrice *  
    d.Quantity) as Total_Spend  
    from customers c  
    join orders o on c.CustomerID = o.CustomerID  
    join `order details` d on o.OrderID = d.OrderID  
    group by c.CustomerID),  
preferred_category as (  
    select c.CustomerID, s.CategoryName, sum(d.Quantity) as  
    Total_Quantithy, count(d.ProductID) as CategoryPurchases,  
    rank() over(partition by CustomerID order by sum(d.Quantity) desc) as category_rank  
    from customers c  
    join orders o on c.CustomerID = o.CustomerID  
    join `order details` d on o.OrderID = d.OrderID  
    join products p on d.ProductID = p.ProductID  
    join categories s on p.CategoryID = s.CategoryID  
    group by c.CustomerID, s.CategoryName)  
    select cs.CustomerID, cs.Order_Count, cs.Total_Spend,  
    pc.CategoryName, CategoryPurchases  
    from customer_summary cs  
    join preferred_category pc on cs.CustomerID = pc.CustomerID and pc.category_rank =  
    1;
```

- **Excel EDA (Pivot Analysis)**

The query results were exported into Excel and analyzed using pivot tables and pivot charts.

- **Pivot Table:** Summarized Order Count, Total Spend, and Category Purchases per customer.
- **Pivot Chart:** Visualized clusters of customers, comparing spend, frequency, and category preference.
- **Slicers:** Enabled dynamic filtering by Product Category and Category Purchases for interactive analysis.

CustomerID	Sum of Order_Count	Sum of Total_Spend	Sum of CategoryPurchases
ALFKI	6	4596.2	4
ANATR	4	1402.95	5
ANTON	7	7515.35	5
AROUT	13	13806.5	5
BERGS	18	26968.15	13
BLAUS	7	3239.8	4
BLONP	11	19088	5
BOLID	3	5297.8	2
BONAP	17	23850.95	10
BOTTM	14	22607.7	6
BSBEV	10	6089.9	3
CACTU	6	1814.8	5
CENTC	1	100.8	1
CHOPS	8	12886.3	5
COMMAL	5	2810.75	4

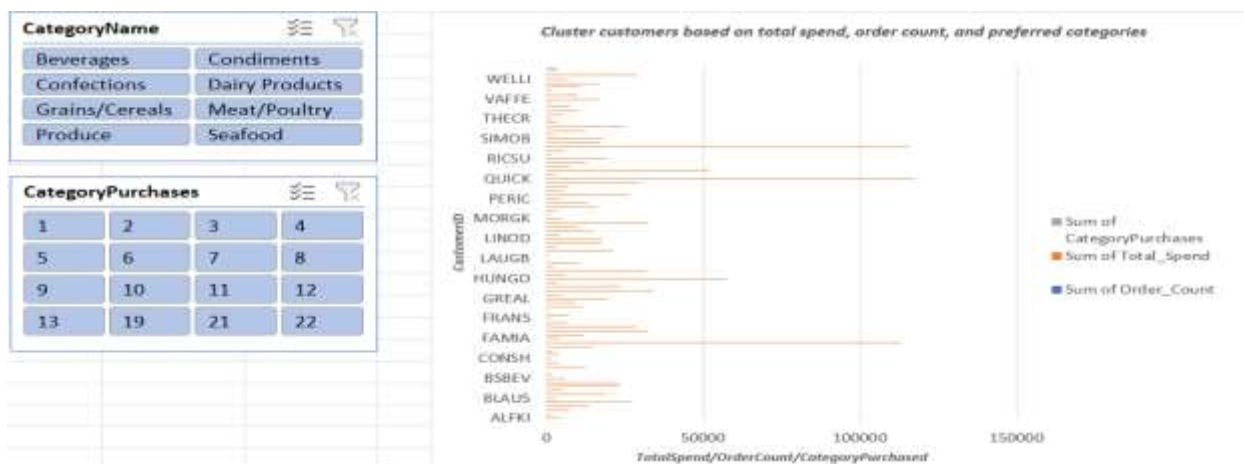


Fig1: Pivot table and chart showing clustering of customers (High, Medium, Low spend groups).

Key Insights

- **High-Value Customers** (e.g., QUICK, SAVEA, ERNSH)
 - Very high total spend and consistent order counts.
 - Strong preference for premium categories like Seafood and Dairy Products.
 - Can be classified as *loyal customers*.
- **Medium-Value Customers**
 - Moderate spending and fewer purchases.
 - Still show consistent preferences (e.g., Beverages, Confections).
 - Represent *growth opportunities* if targeted with promotions.
- **Low-Value Customers**
 - Small spend and low order count, often spread across multiple categories.
 - Less loyalty, tend to be occasional or *price-sensitive buyers*.

Conclusion

Customer segmentation reveals clear **clusters**:

- High-value customers drive most of the revenue and should be rewarded with loyalty programs.
- Medium-value customers are the best candidates for cross-sell/upsell campaigns.
- Low-value customers require targeted promotions/discounts to improve engagement.

Overall, clustering based on spend, order frequency, and category preference provides a strong foundation for customer segmentation, personalized marketing, and revenue growth.

(Similar analysis was done for all 15 business questions, covering sales performance, top products, customer geography, employee performance, inventory, and shippers.)

2. Power BI Analysis (15 Business Questions)

A robust star schema model was designed to integrate multiple datasets for analysis:

- **Fact Table:**

- **Order Details** → Contains transaction-level data (OrderID, ProductID, UnitPrice, Quantity, Discount).
- **Dimension Tables:**
 - Orders → Links customers, employees, shippers, and order dates.
 - Customers → Provides customer demographics (location, company details).
 - Employees → Employee details and reporting structure.
 - Products → Product catalog with prices, stock levels, and categories.
 - Categories → Classification of products into groups.
 - Suppliers → Supplier details linked to products.
 - Shippers → Information about shipping companies.

Relationships

- Orders → Customers (CustomerID)
- Orders → Employees (EmployeeID)
- Orders → Shippers (ShipperID)
- Order Details → Products (ProductID)
- Products → Categories (CategoryID)
- Products → Suppliers (SupplierID)

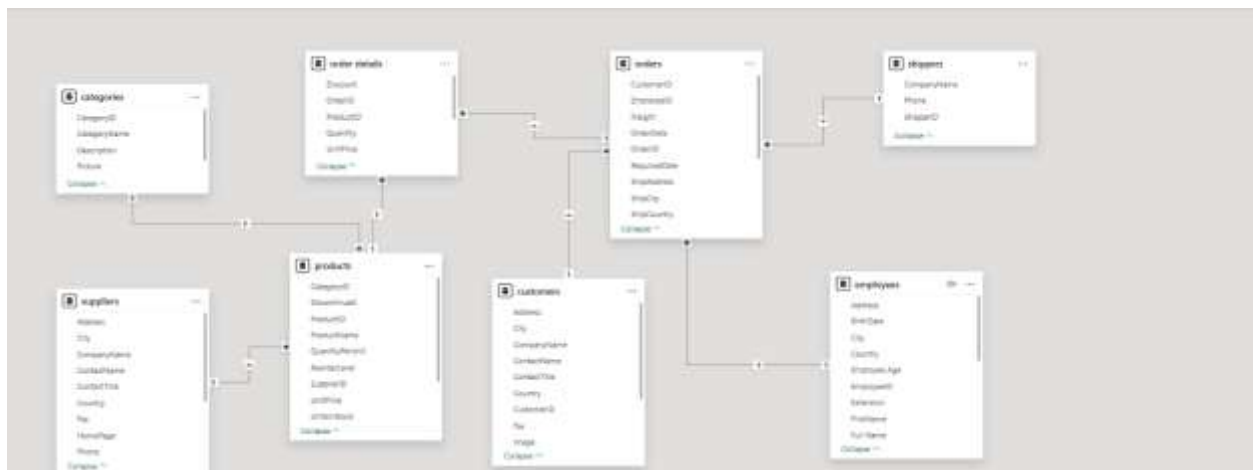


Fig2: Power BI Data Model view showing star schema relationships.

Example: (How does customer distribution vary across different countries or cities?)

Business Question

How are customers distributed across different countries and cities, and which regions contribute the most to the customer base?

Power BI Visualization

- **Bar Chart:** Total Customers by Country.
- **Map Visual:** Global view of customer locations.
- **Card Visuals:**
 - Total Customers = 91
 - Countries Covered = 21

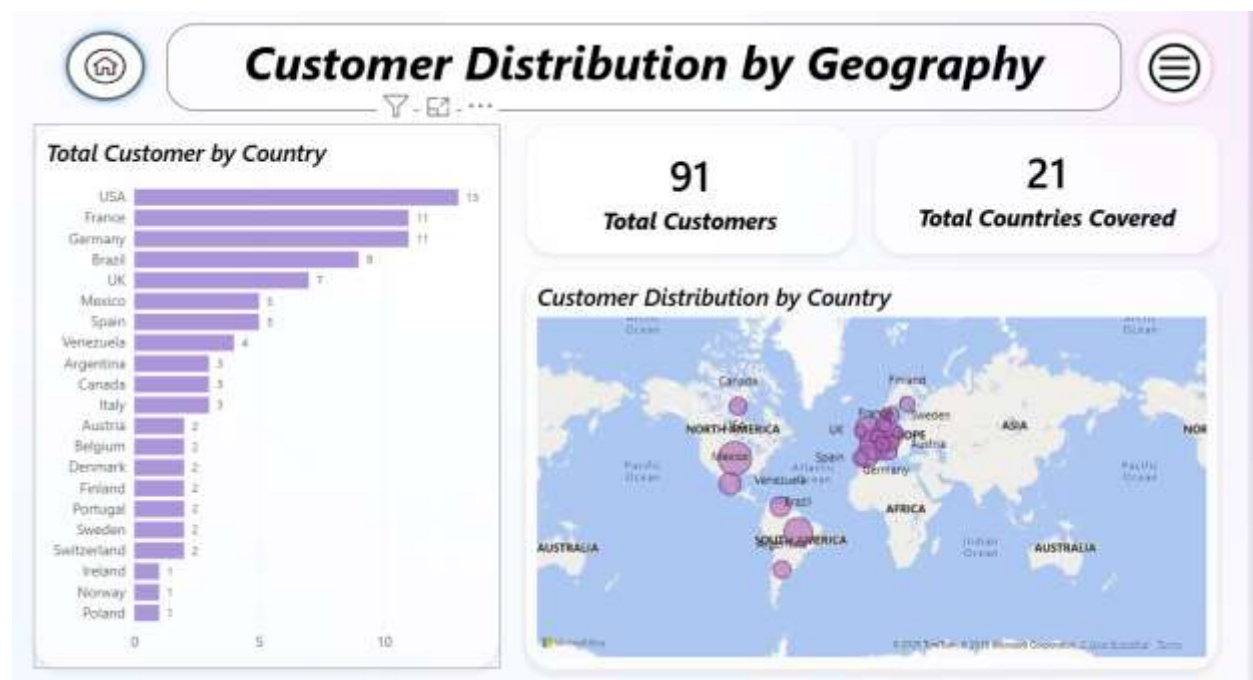


Fig3: Power BI report showing customer distribution by geography.

DAX Measure

Customer Count = DISTINCTCOUNT(Customers[CustomerID])

This measure counts unique customers, used in both cards and charts.

Key Insights

- USA has the highest customer base (13), followed by France and Germany (11 each).
- Customers are distributed across 21 countries, showing strong global coverage.
- Europe and North America are the most concentrated markets, while countries like Norway, Ireland, and Poland each have only one customer.

Conclusion

The analysis reveals that a majority of Northwind's customers are concentrated in Europe and North America, with the USA as the single largest market.

This insight helps the company:

- Focus marketing and sales campaigns on high-density countries.
- Explore expansion opportunities in underrepresented regions.
- Align inventory and logistics to serve dominant markets more effectively.

(Similar analysis was done for all 15 business questions)

3. Dashboard



Fig4: Sales Performance Dashboard

After completing the 15 business questions individually, the insights were consolidated into a comprehensive Sales Performance Dashboard. This dashboard provides a holistic view of Northwind Traders' operations in a single page, enabling quick decision-making.

Dashboard Features

- **KPI Cards:**
 - Total Sales: 1.27M
 - Total Orders: 830
 - Total Categories: 8
 - Total Customers: 91
 - Top Performer: *Margaret Peacock* (Employee)

- **Top Product by Revenue**
 - Products like *Côte de Blaye* and *Thüringer Rostbratwurst* emerge as the most revenue-generating items.

- **Sales Trend Over Time**
 - Monthly sales trend shows seasonal fluctuations, with a peak around **June**.

- **Supplier Contribution Analysis**
 - Highlights supplier performance, identifying those contributing most to the product supply chain.

- **Sales by Category**
 - Beverages, Dairy Products, and Confections lead in sales contribution.

- **Customer Geography (Map View)**
 - Displays global customer distribution, concentrated heavily in **Europe and North America**.

Key Insights from Dashboard

1. **Revenue Concentration:** A few products and categories dominate total sales, suggesting cross-selling opportunities.

2. **Employee Performance:** Top employees like *Margaret Peacock* drive significant sales, indicating potential for recognition programs.
3. **Regional Focus:** North America and Europe are the strongest markets, while other regions are underrepresented.
4. **Supplier Impact:** A small set of suppliers contribute to most of the inventory, highlighting dependency risks.

Conclusion

The **Sales Performance Dashboard** transforms individual insights into a **strategic overview**. By combining KPIs, top product/customer analysis, supplier performance, and geography into one view, stakeholders can:

- Monitor overall sales health at a glance.
- Identify strengths and weaknesses across dimensions (products, customers, employees, suppliers).
- Make **data-driven decisions quickly** without navigating multiple report pages.

CONCLUSION

The Northwind Traders Sales Analytics Project successfully demonstrated the end-to-end process of transforming raw data into meaningful business insights using Excel, MySQL, and Power BI.

- Through SQL queries, we extracted and cleaned data from multiple relational tables, ensuring a reliable analytical foundation.
- Using Excel, we conducted exploratory data analysis (EDA) with pivot tables, charts, and slicers to address initial business questions and derive actionable insights.
- In Power BI, we integrated data directly from MySQL, performed data modeling (fact–dimension schema), and answered 15 key business questions through interactive visualizations.
- Finally, we consolidated findings into a Sales Performance Dashboard, providing a single-page strategic view of company performance across sales, customers, products, employees, and suppliers.

The analysis revealed:

1. **High-value customers and products** drive the majority of revenue.
2. **Employee performance** varies significantly, with top performers contributing disproportionately.
3. **Regional sales concentration** highlights untapped growth opportunities outside Europe and North America.
4. **Supplier dependencies** indicate risks but also opportunities for renegotiation or diversification.

This project not only provided descriptive analytics (what happened) but also enabled diagnostic insights (why it happened), laying the groundwork for future predictive and prescriptive analytics initiatives.

Overall, the project demonstrates how structured data analysis can empower Northwind Traders' stakeholders to make data-driven decisions, improve operational efficiency, strengthen customer relationships, and enhance competitiveness in the wholesale trading market.